Experiment No. 8: Image Handling and Dilation

Aim: To read an image in Python and perform image dilation using the dilate() function in OpenCV.

Program (Google Colab):

```
import cv2

import matplotlib.pyplot as plt

from google.colab import files


# Upload an image

uploaded = files.upload()

image_path = next(iter(uploaded))


# Read the image

img = cv2.imread(image_path)


# Convert BGR to RGB

img = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)


# Create a kernel

kernel = cv2.getStructuringElement(cv2.MORPH_RECT, (5, 5))


# Perform dilation

dilated = cv2.dilate(img, kernel, iterations=1)


# Display original and dilated images

plt.figure(figsize=(10, 5))
```

```
plt.subplot(1, 2, 1)
plt.imshow(img)
plt.title("Original Image")
plt.axis("off")

plt.subplot(1, 2, 2)
plt.imshow(dilated)
plt.title("Dilated Image")
plt.axis("off")

plt.show()
```

OUTPUT:



Expected Output:

1. Original image.
2. Dilated image with thicker and expanded foreground regions.

Result: The image was successfully read using OpenCV, and image dilation was performed using the cv2.dilate() function in Google Colab.